

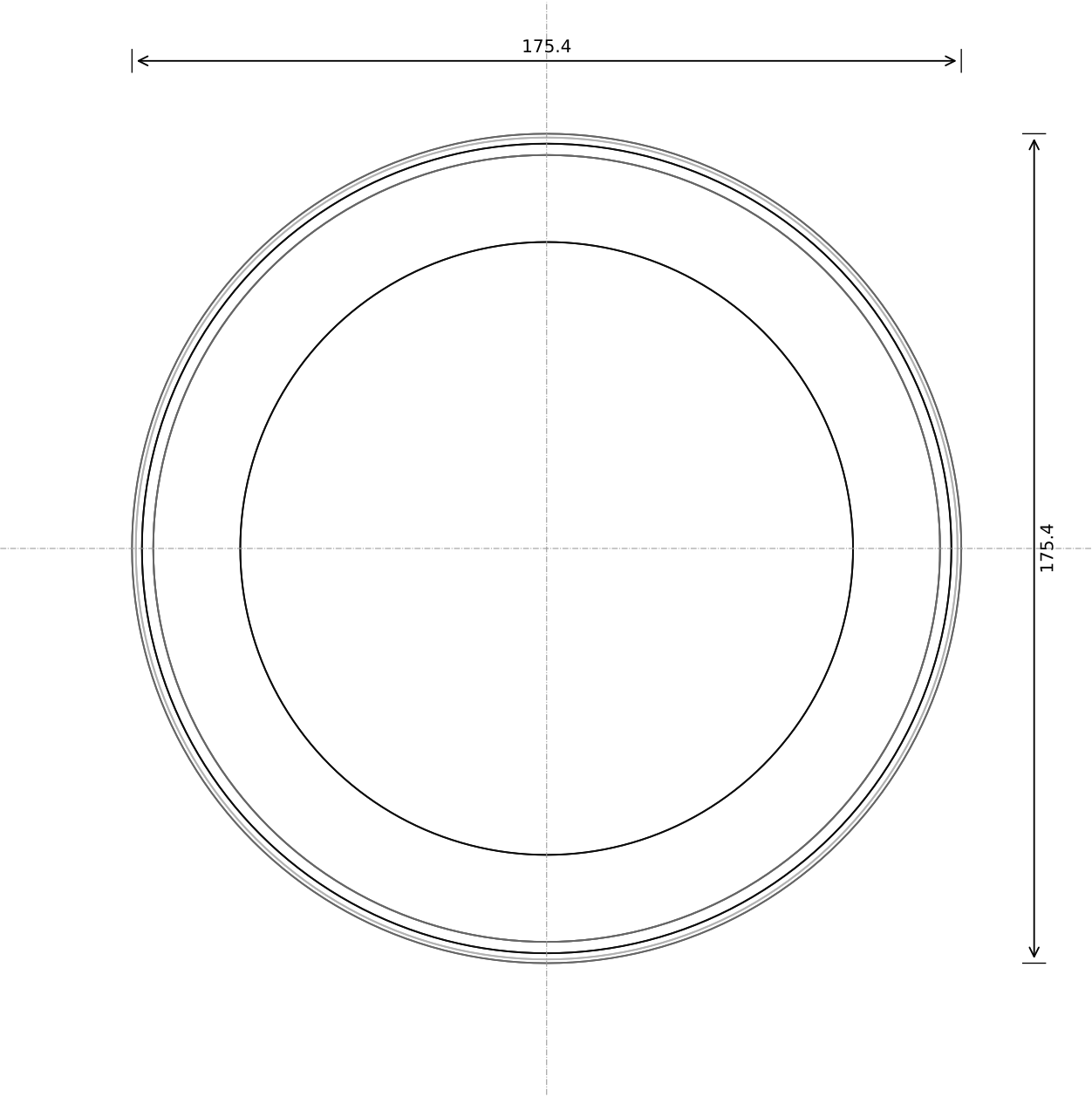
the 60 — v12 — sheet 0: the ASSUMED numbers (every one of these must be verified on the real part before the drawing is trusted)

2026-09-06. Stack: plate 8.0, adapter 14.78-16.38 (socket to 23.68), code band 17.4-21.6, seat 28.0, glass 28.50-30.48, lens to 32.98, knob top 36.38, overall 37.88 with the pad. Ø175.

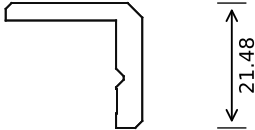
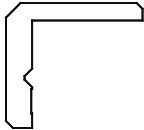
1. cover lens: Ø140.0 x 2.5	specified separately; carrier rim and lip overlap designed to it
2. bonding tape: 0.5	double-sided foam tape between the seat and the glass edge (RULING 5 Sep: glue allowed for the display)
3. panel back component area: 60.0 x 68.36 x 1.5 at -14.0	position from the drawing's back view
4. Active Cooler position: from Pi x 2.5, fan from x 43.0	no CAD body; footprint and fan position from photographs
5. cooler base above the board: 2.8	SoC 2.4 + thermal pad
6. fan intake plenum: 8.0	minimum free height above the blower
7. micro-HDMI receptacles: x 26.0/39.5	not in the vendor STEP; drawn as envelopes
8. HDMI ribbon plugs: 12x8x5 micro end, 21x12x6 A end	flat FPC HDMI cable, 200 mm
9. adapter board thickness and part heights: 1.6; parts 2.5, underside 1.0, flex connector 2.0	outline, holes, socket size and position PUBLISHED (datasheet drawing)
10. display connect board: 30.0 x 46.0 x 1.2, connectors 2.0	part of the adapter kit; size scaled from its photograph, hole positions assumed at the corners
11. Pi standoffs and washers: M2.5 x 12.0 hex, 0.5 washers	the adapter stack: closing-plate land / washer / Pi / standoff / adapter
12. USB-C up-angle plug: 12 x 12.5 x 6.5 head	measure the lead
13. audio / motion / touch / converter boards: envelopes	electrical layout not done
14. barrel jack body: 14.4 x 9.0 x 11.0 on a vertical board	PJ-063AH; CAD body to fetch
15. right-angle barrel plug: Ø8.5 body	measure the brick's plug
16. MOTOR_BASE_OD / MOTOR_BASE_H: 30.0 / 4.3	carried from v9
17. MOTOR_BAND_T: 0.6	carried from v9
18. servo pushrod: 3 mm bridge to the tab	carried from v9; servo on the plate in a frame
19. LED strip: 5.0 wide x 1.6 thick	any 5 mm addressable strip; a wider strip raises the halo band

Build-time changes against V11-SPECIFICATION.md are listed in V11.md. Sheets 1-11 follow: one per made part.

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

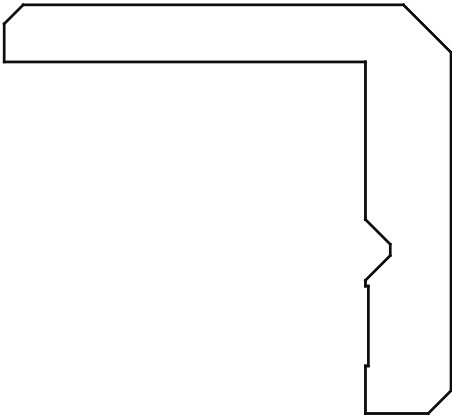


SECTION A-A (front, plane y = 0; x across, z up)



21.48

DETAIL C (section A-A, x 58 to 90, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- outside $\varnothing 175.4$, bore $\varnothing 166.4$ (0.45 to the panel's chin corners); wall 4.5
- top rim: lip inner edge $\varnothing 128.4$, 1.0 inner chamfer, 20 flat, 2.5 outer chamfer (RULING 6 Sep: rim ≥ 20); lip 3.0 thick, top at z 36.38
- outer chamfer 2.5 x 45° smooth; inner chamfer 1.0; skirt chamfer 1.2
- V-groove 90°, root r 84.50399999999999, z 21.9-25.1
- code-band recess 0.15 deep, z 17.4-21.6
- diamond knurl 78 starts x 2, 1.0 deep, helix 33.2°, 3 whole rows z 16.9-33.1; skirt bottom z 14.9
- wall under the knurl root 3.5

the 60 — v12

Knob

Cadrane · sheet 1 of 13 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / CNC 6082-T6, hard anodised (production)

Process: print top face down, 0.16 mm; no supports

mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,

static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

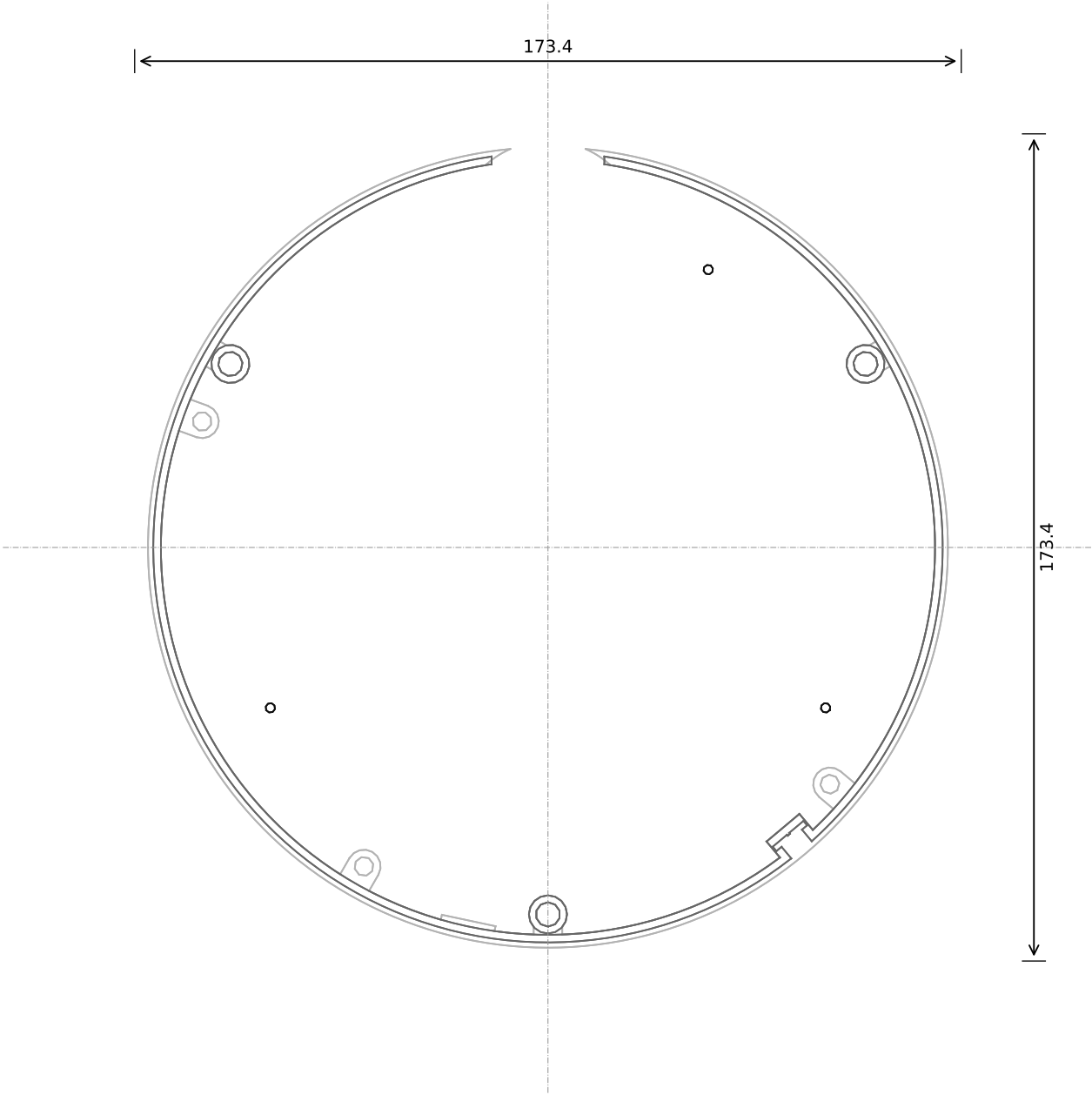
Part: knob_body

Overall 175.4 x 175.4 x 21.5

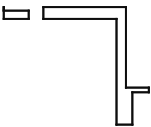
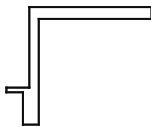
Volume 74.36 cm³

Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

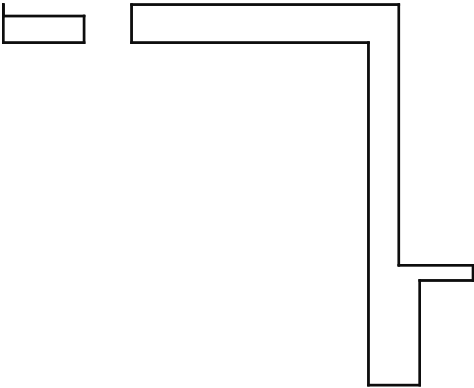


SECTION A-A (front, plane y = 0; x across, z up)



22.19

DETAIL C (section A-A, x 57 to 89, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



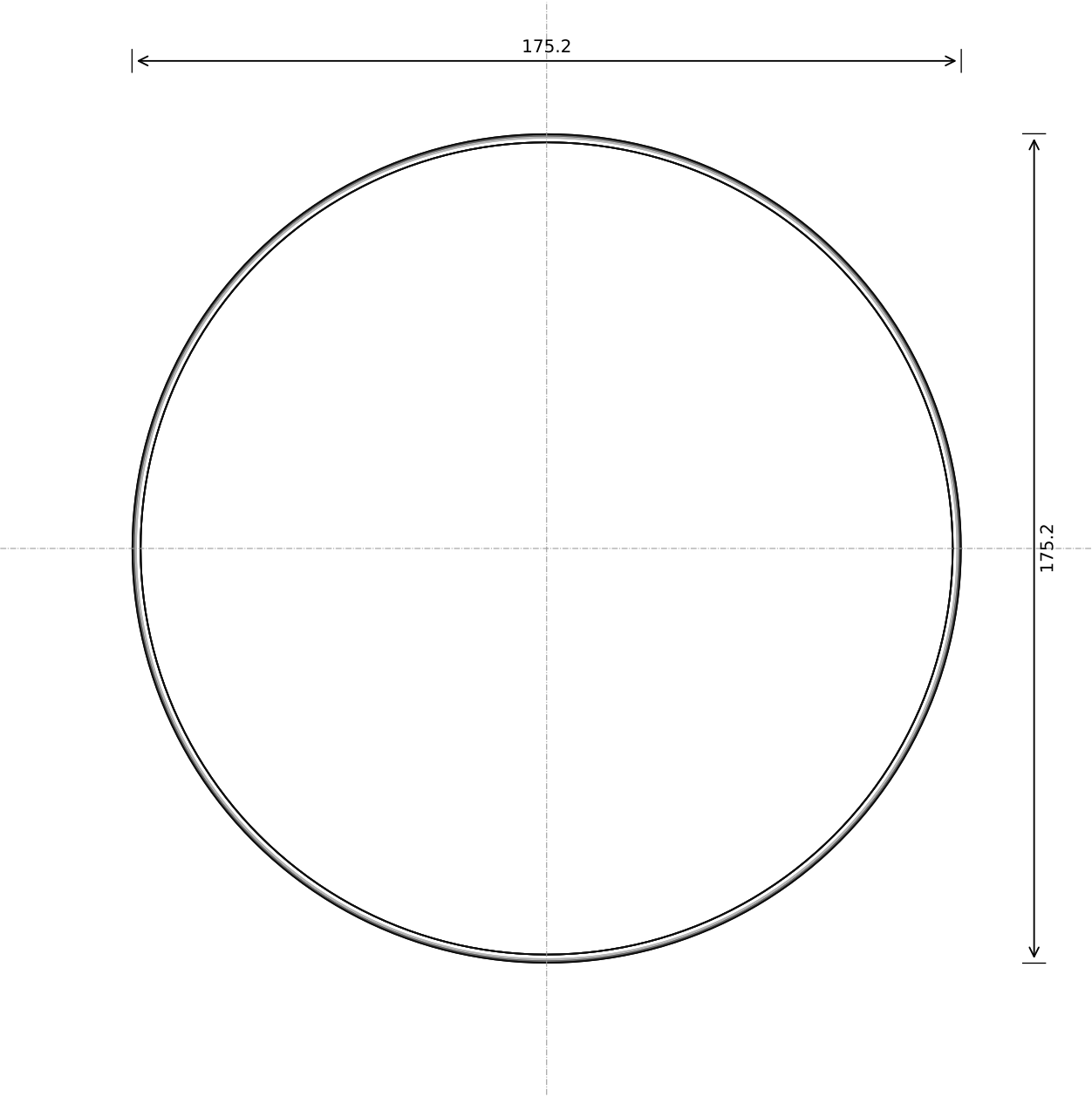
.

Key dimensions (from the parameter block):

- wall r 81.204-82.80399999999999, z 8.0-28.0
- seat flange r 62.0-82.80399999999999, z 26.0-28.0; three Ø2 locating nubs at [60, 210, 330]°
- LED-strip band r 82.80399999999999-83.90399999999998, z 8.0-13.5 (strip r 84.00399999999998-85.60399999999997); diffuser lip r 82.80399999999999-86.704, z 13.5-14.3
- wheel posts Ø8.0 at r 76.93, az [30, 150, 270], to z 20.9; bush bores Ø5.1; wall windows 15.2 wide, open to the top from z 20.6; the seat flange is cut away above each wheel
- pillars Ø7 at r 77.204, az [160, 240, 320], to z 16.0; M3 inserts from below (the plate screws) - nothing screws in from above
- encoder tower at 310.0° to z 24.0, board slot open to the top, lead notch through its inner face at z 15.5-16.8, window through the wall at the code band; motor window open to the seat; LRA pad at 258.0°
- flex slot 32 x 2.5 through the seat at 0°, offset -4; 97 vent slits 1.0 x 3.1 at z 21.9-25.0, 2.5° grid
- motor relief at 90°

the 60 — v12		Internal structure	
Cadrane · sheet 2 of 13 · 2026-09-06 · drawn from src/params.py + model.py (build123d)			
4.3 Material: PETG (prototype) / PA12 (production)		Part: internal_structure	
Process: print upright on the wall foot, 0.20 mm; supports ONLY under the seat flange (inside) and the diffuser lip (outside)		Overall dimensions: 173.4 x 173.4 x 28.0	
to the seat: LBA pad at 258.0° mm: tolerance ±0.1 unless stated; clearances: running 0.30,		Volume 37.49 cm³	
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.		Scale: see each view (fit)	

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 58 to 90, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. r 85.8-86.7 at the bottom, to r 87.7 at the top; z 8.0-13.5
- 2. 0.5 chamfer on the outside top edge
- 3. retained by the structure's lip (r to 86.7); sits on the rim ring 0.2 outside the LED strip

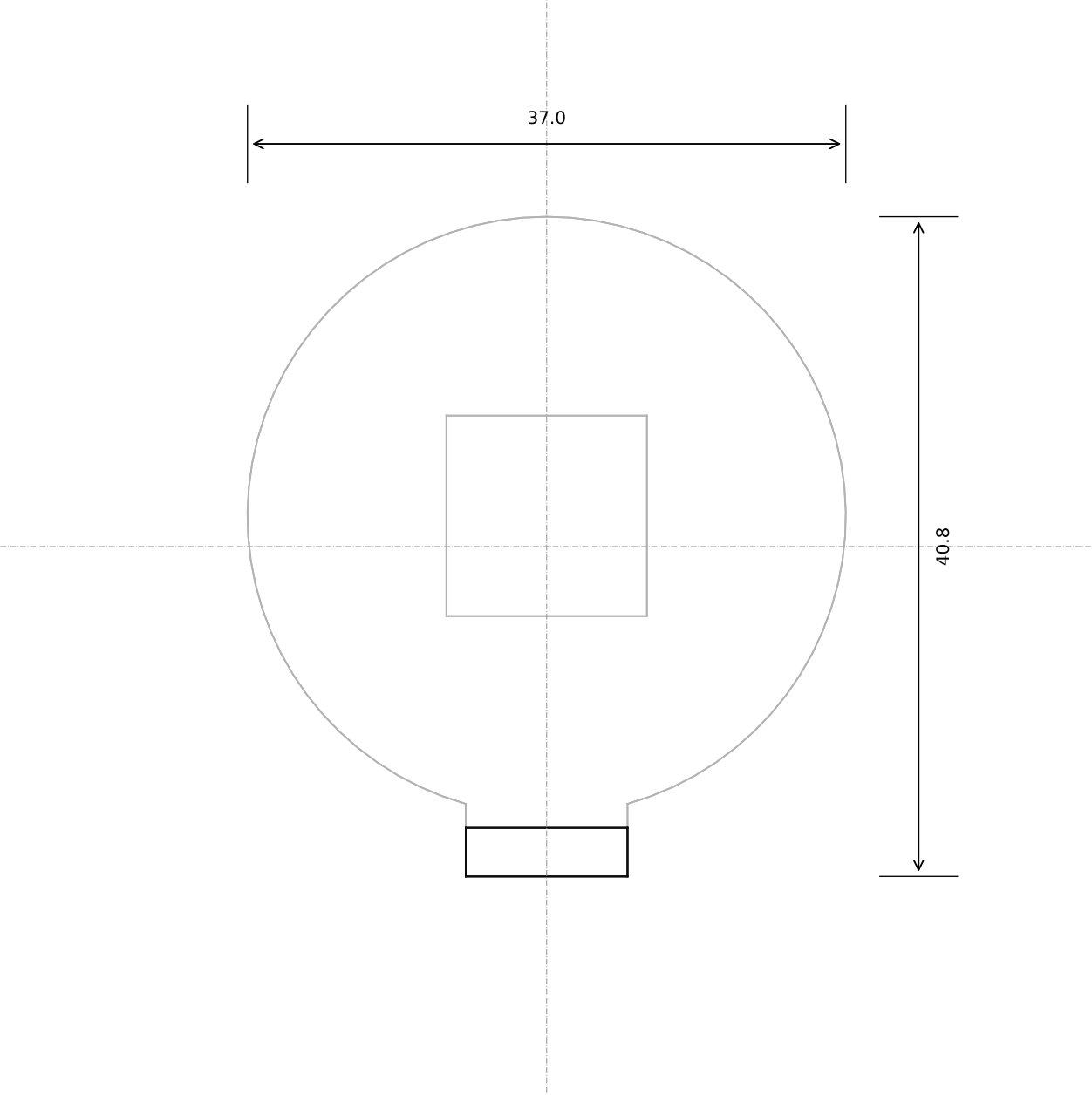
the 60 — v12Halo diffuser

Cadrane · sheet 3 of 13 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

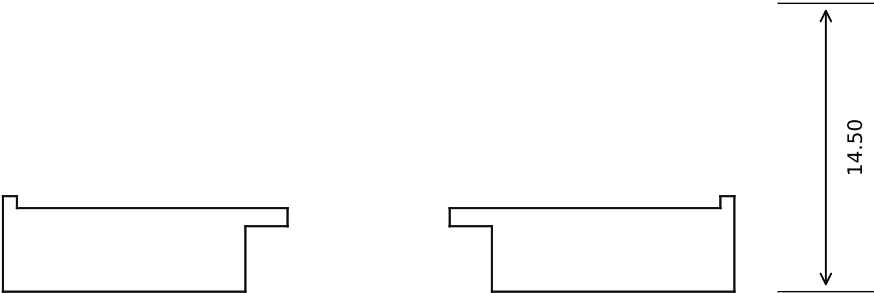
Material: opal PMMA (production) / natural PETG (prototype)
Process: print on its underside, 0.20 mm; no supports
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: halo_diffuser
Overall 175.2 x 175.2 x 5.5
Volume 4.12 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

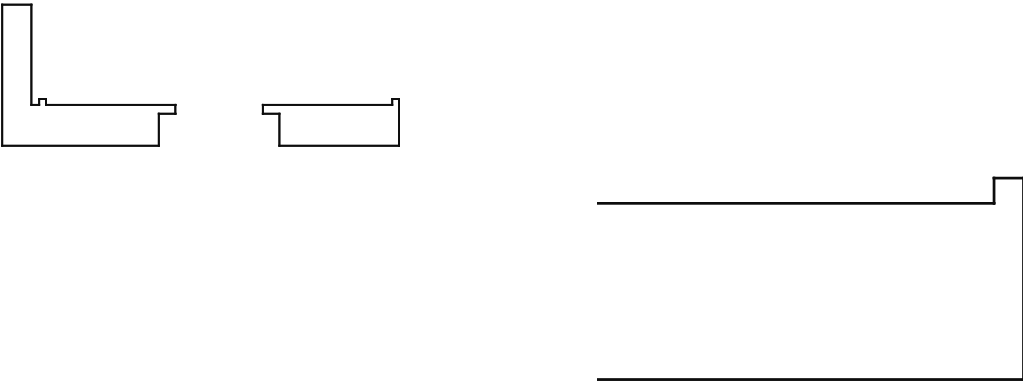


SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 9 to 20, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. shoe Ø37 x 4.2; locating rim Ø35.6/37 x 0.6
- 2. commutation-board pocket 12.4 x 12.4 x 3.3; sight hole Ø9
- 3. push tab 3.0 x 10.0 to z 14.5 (v11: 28.5 through the deck; the servo is now beside it on the plate)
- 4. travel 2.4 radial in the plate's stadium hole

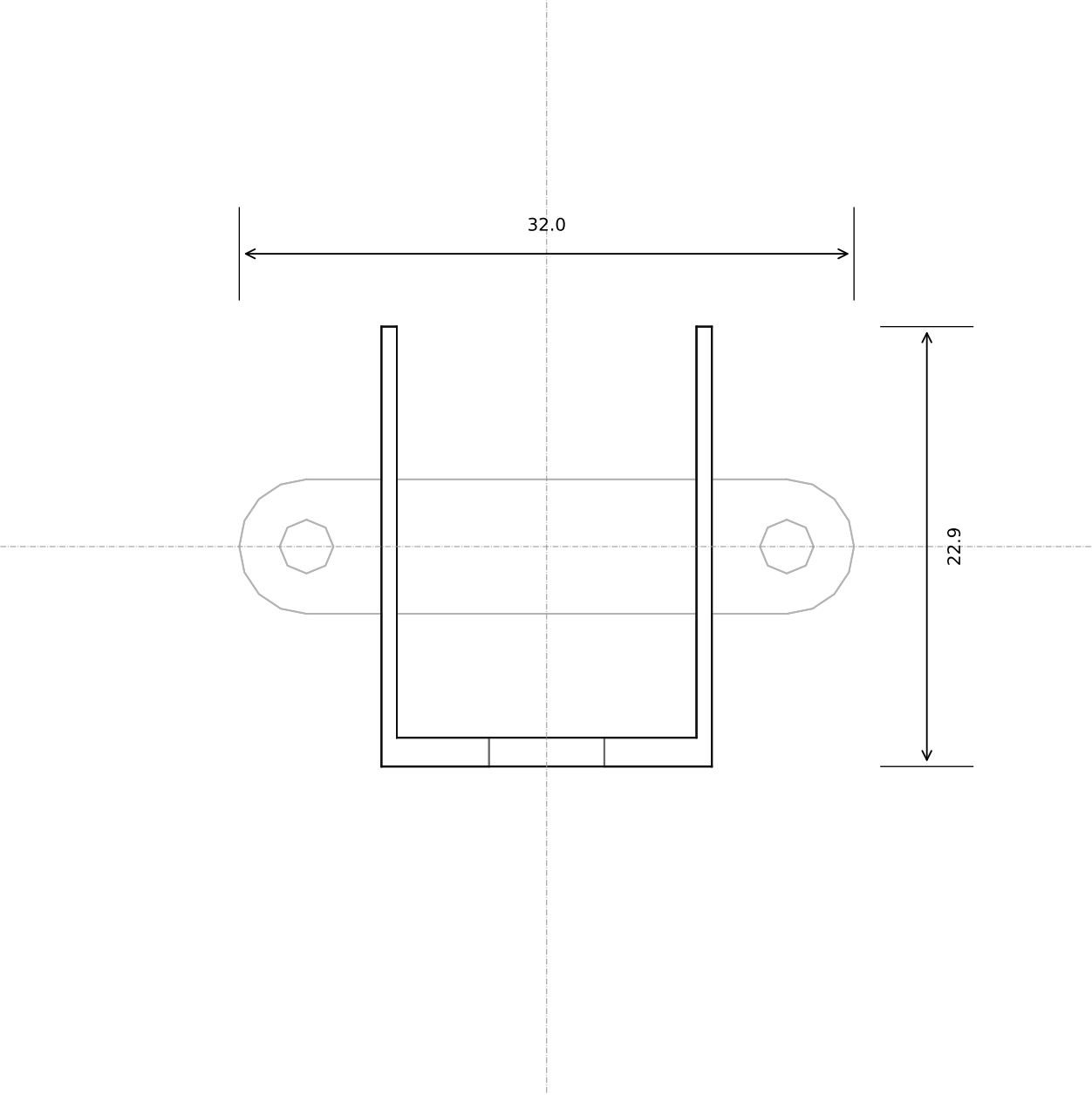
the 60 — v12

Motor carriage

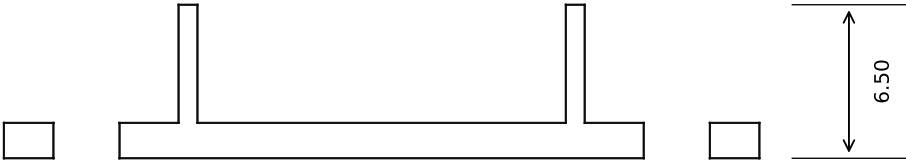
Cadrane · sheet 4 of 13 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG	Part: carriage
Process: print standing on the shoe, 0.16 mm; no supports (12 mm bridge over the 37 mm Ø hole)	Overall 37 mm x 40.8 mm x 14.5 perimeters for the tab
mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Volume 4.48 cm³
	Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

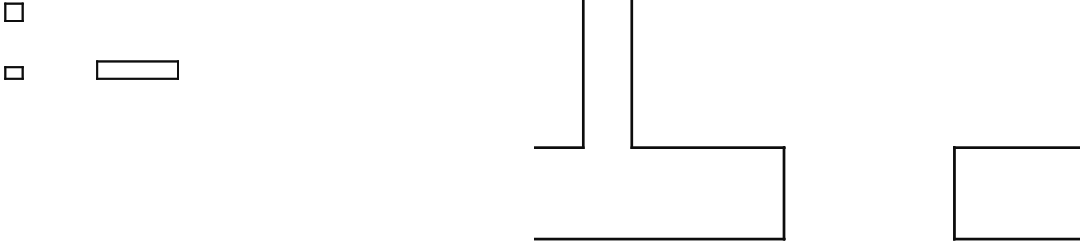


SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 8 to 18, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. floorless frame round the AGFRC envelope 21.4 x 15.2 x 6.0; walls 1.0, 6.5 tall; the servo lies on the plate at r 19.0-40.4, az 90
- 2. open toward the carriage tab; lead notch in the inner end wall
- 3. 2 ears at t ±12.5, M2.5 x 4 into tapped blind holes in the plate's web

the 60 — v12

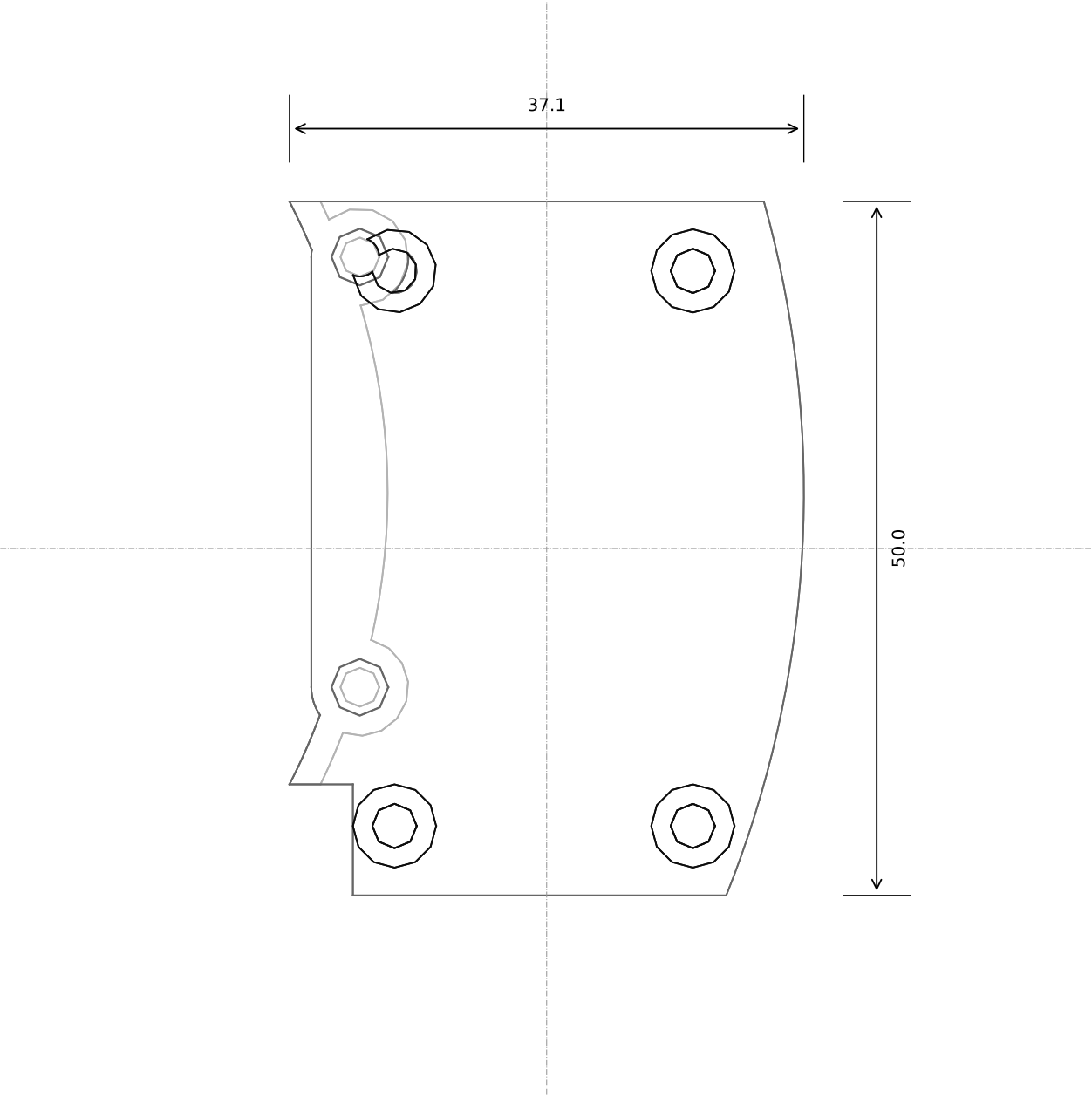
Servo frame (on the plate)

Cadrane · sheet 5 of 13 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

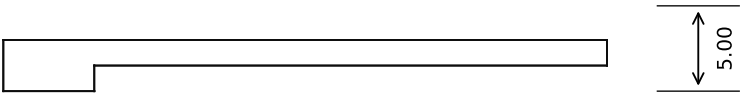
Material: PETG
Process: print flat, 0.20 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: servo_mount
Overall 32.0 x 22.9 x 6.5
Volume 0.64 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 9 to 21, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



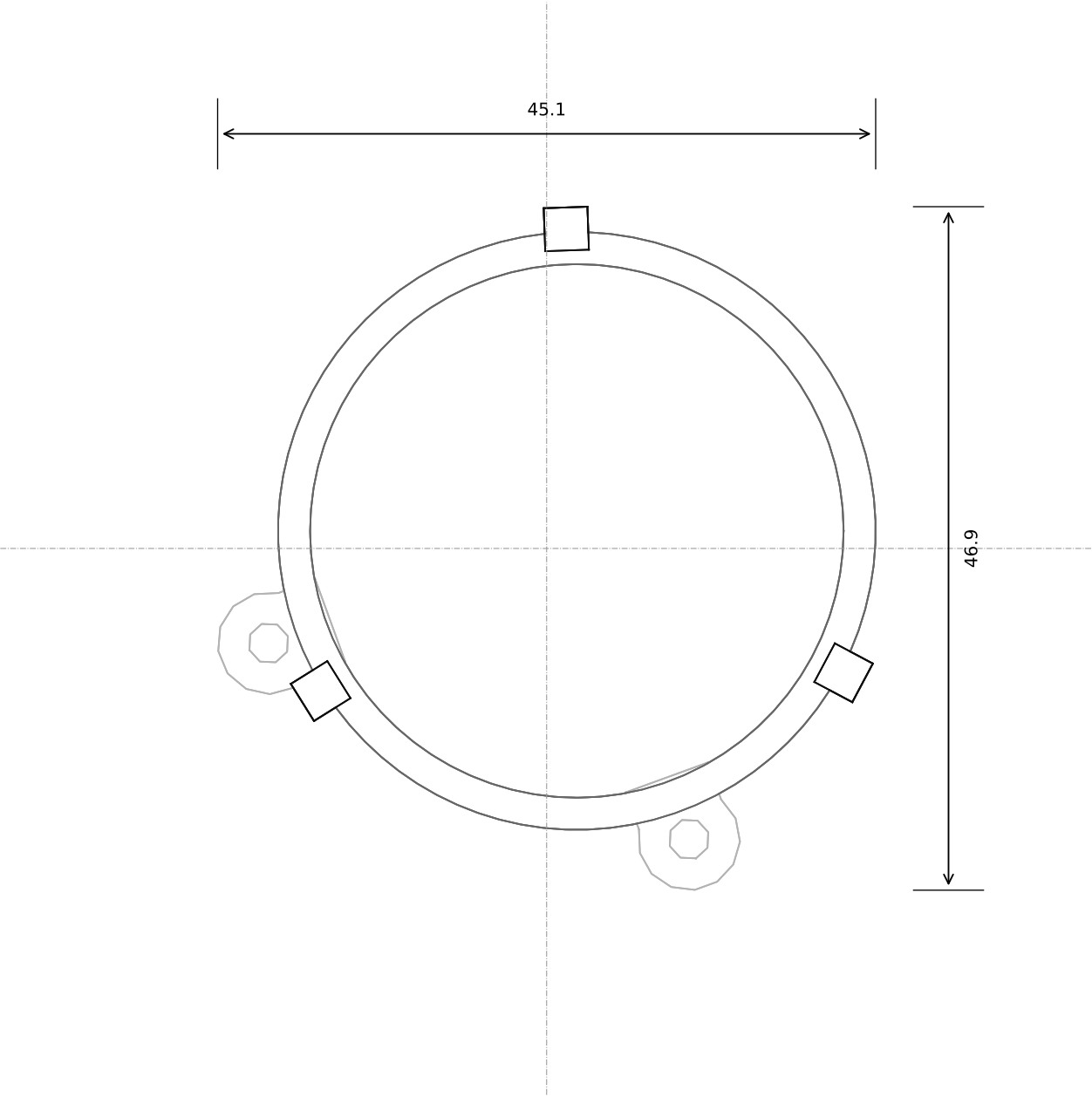
- Key dimensions (from the parameter block):**
- 1. slab 1.5 thick at z 9.5-11.0, r 46-78, t -29..21, bridging the port slot; its outer end rests on the port face's rail (z 9.5)
 - 2. foot strip r 46-48 on the plate; two csk ears at [(46.0, 17.0), (46.0, -14.0)], M2.5 into the web
 - 3. four Ø6 bosses 2.0 tall with M2 inserts for the connect board (30.0 x 46.0, holes 3 in from its inner corners, 5.5 from its outer)
 - 4. the -y inner corner cut back to x 45.5 (the audio board's USB plug is there)

the 60 — v12 Connect-board bracket

Cadrane · sheet 6 of 13 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG	Part: connect_bracket
Process: print standing on its straight -y edge (slab vertical, feet and bosses clear by 37, 10 x 200.0 x 50.0 supports)	
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Volume 2.92 cm ³
	Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 11 to 25, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- ring r 18.3-20.5 x 4.45 under the Ø40 flange
- three snap fingers at [92.0, 212.0, 332.0]° from the speaker's centre (device azimuths)
- two ears at r 22.5, [200.0, 290.0]° from the centre; M2.5 x 4 from above into the plate's web (v11: M3 from below)
- speaker centre (-36, 47) - az 127°, r 59.2

the 60 — v12

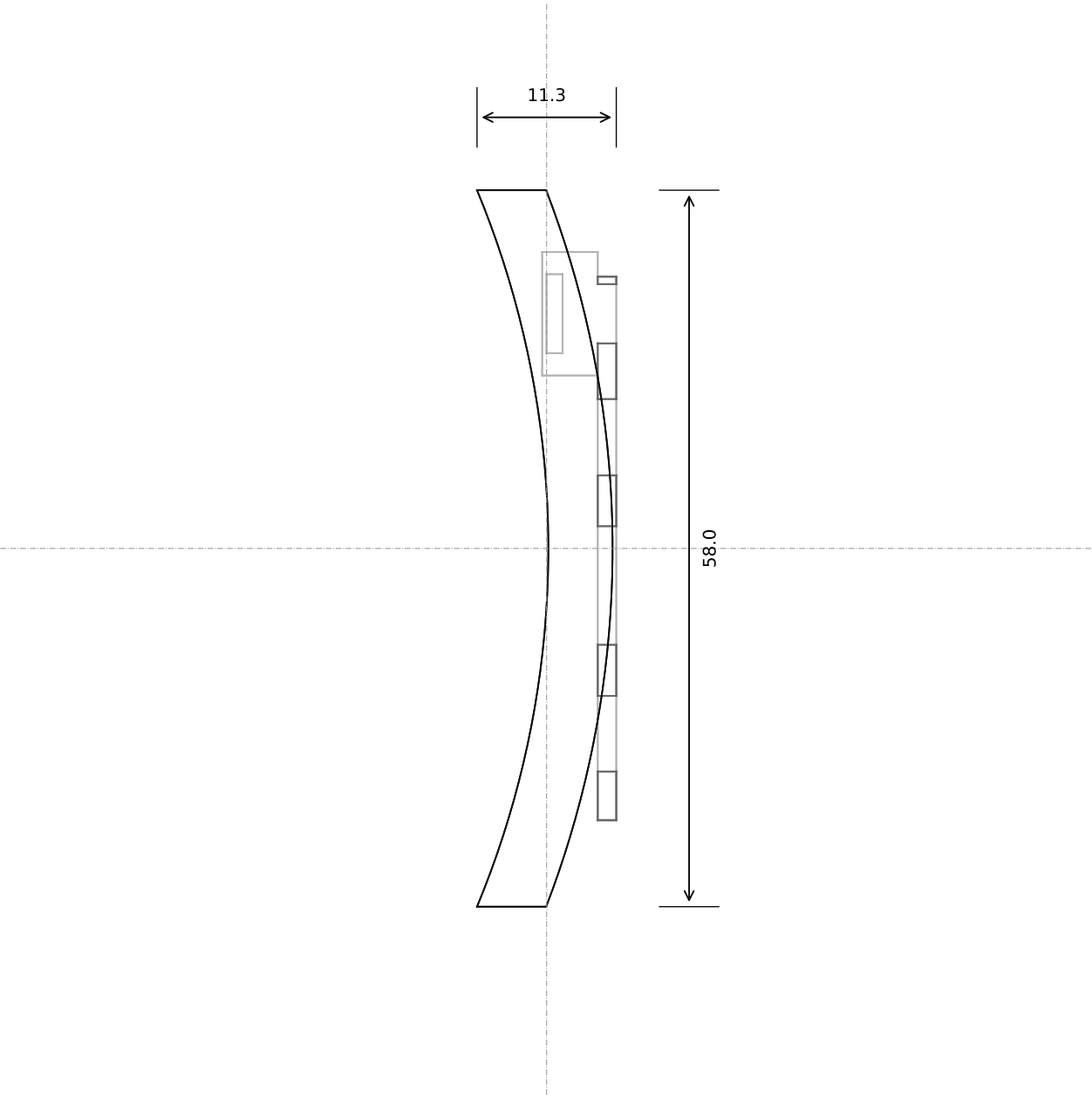
Speaker cradle

Cadrane · sheet 7 of 13 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

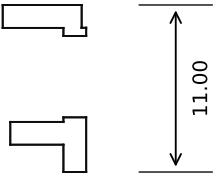
Material: PETG
Process: print flat, 0.20 mm; no supports (fingers full height)
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: speaker_cradle
Overall 45.1 x 46.9 x 7.2
Volume 1.44 cm³
Scale: see each view (fit)

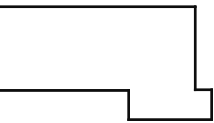
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



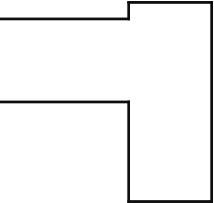
SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 3 to 8, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. 44.0 wide x 1.5 thick at r 79.704, z -1.5-8.0
- 2. barrel nose Ø6.2 at t -15.0, z 3.5; USB-C opening at t -3.0, z 4.5; jack Ø6.3 at t 9.0, z 4.6; light aperture Ø4.8 at t 19.0
- 3. shelf for the USB-C board; foot for the light-sensor board; rail along the top on the plate (1.5 tall, 58 wide, inner edge r 75.7, flat front in the face's plane, corners cut 45° to stay inside the diffuser) with two Ø2.2 holes at r 78.2, t ±26: M2 screws into the plate

the 60 — v12

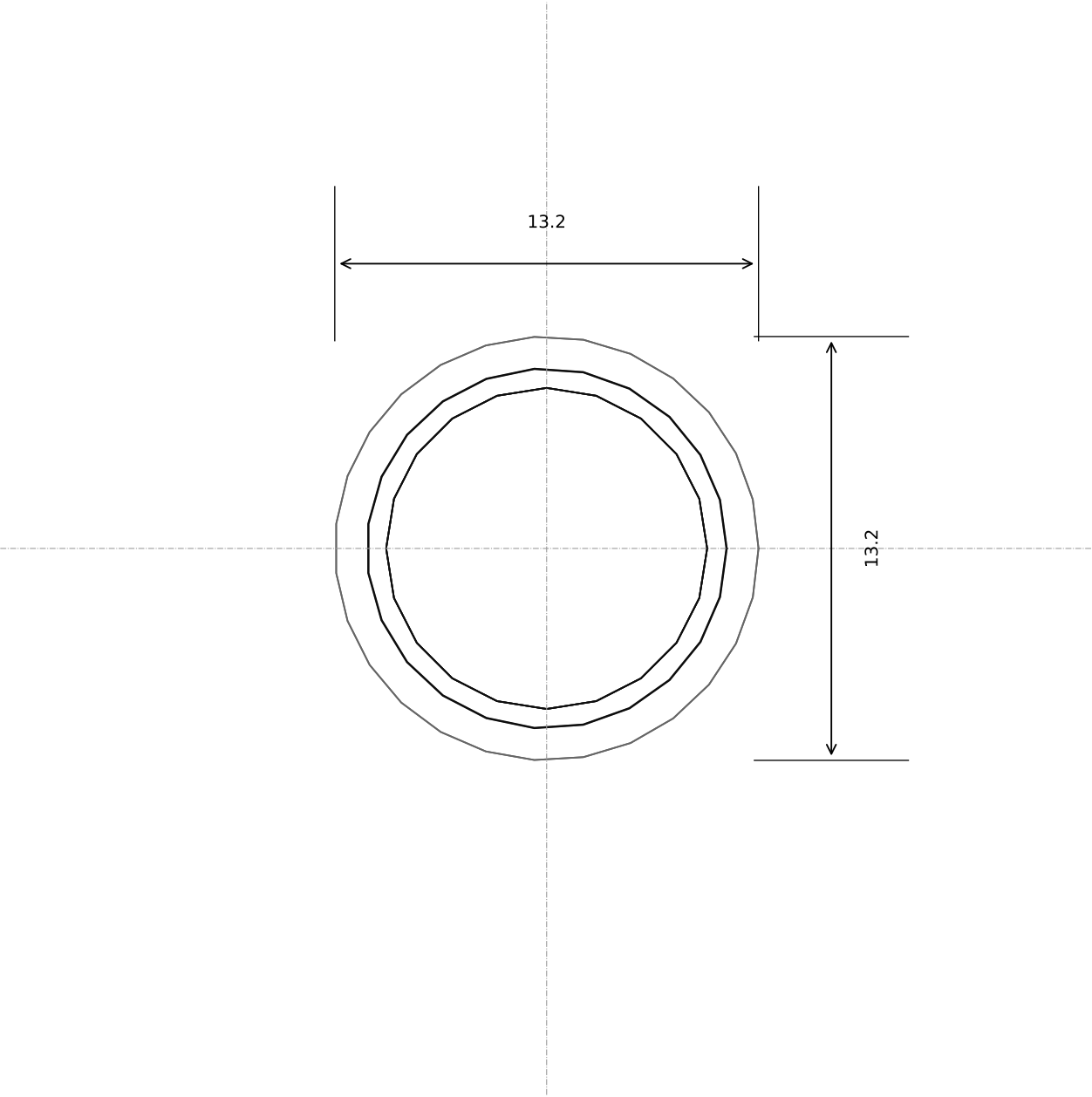
Rear port face

Cadrane · sheet 8 of 13 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG
Process: print lying on its outer face, 0.16 mm; no supports (the socket holes must be printed on the inner face)
mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: port_face
Overall size: 58.0 x 11.0
Volume 1.01 cm³
Scale: see each view (fit)

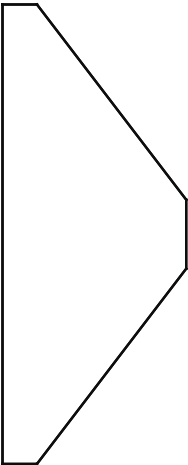
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 3 to 9, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

1. Ø13.2 x 4.0, 90° V 1.3 deep with a 0.6 flat; bore Ø10 press over a 623ZZ

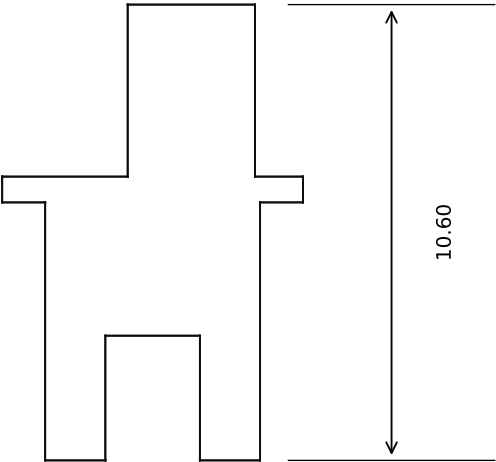
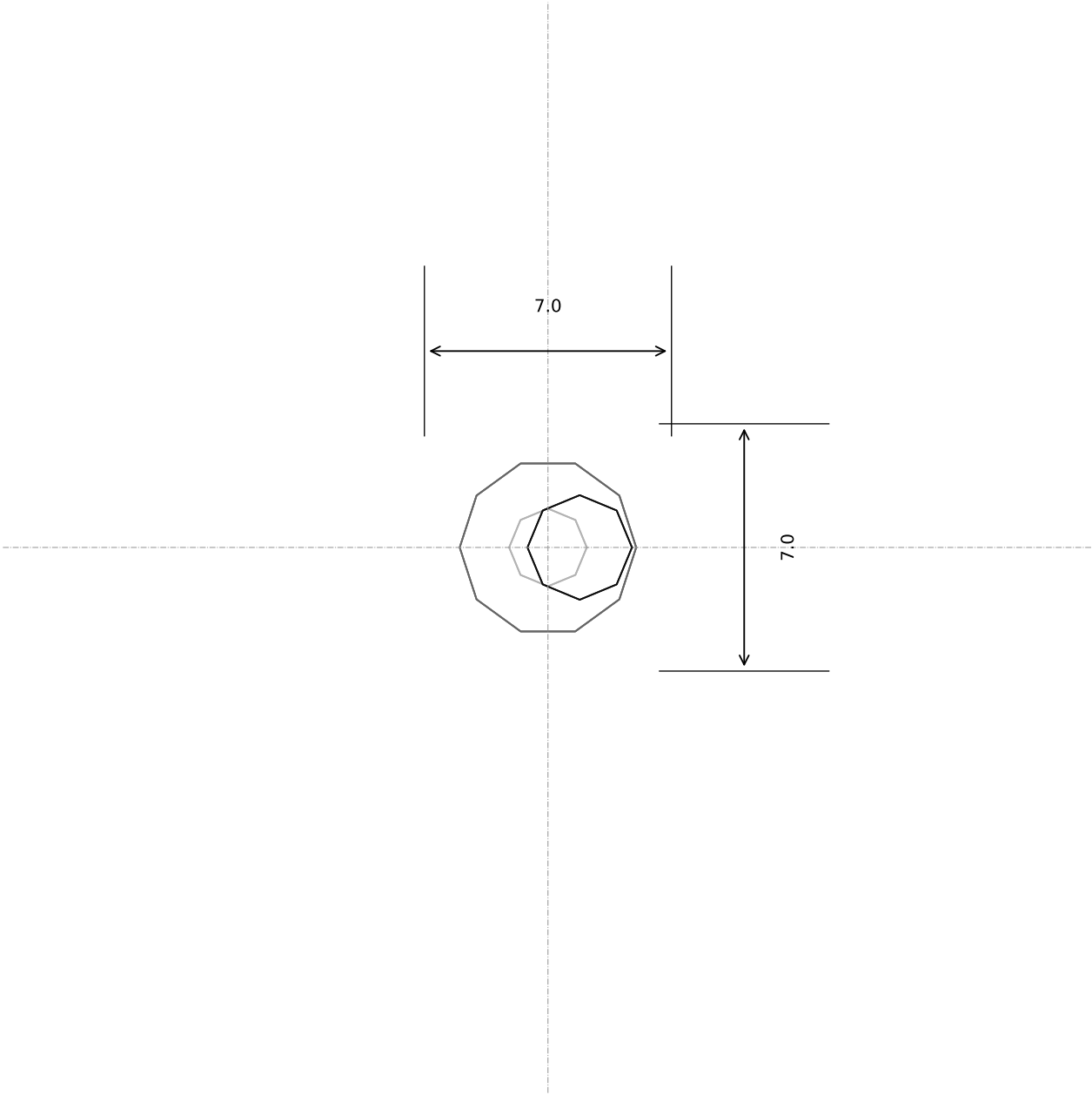
the 60 — v12 Wheel V-collar (x3)

Cadrane · sheet 9 of 13 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / POM (production)
Process: print axis vertical, 0.12 mm
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

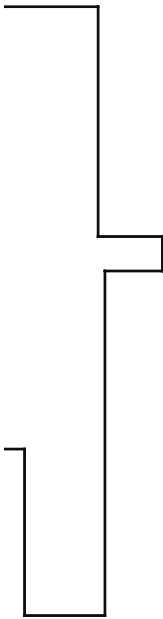
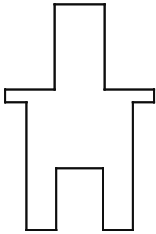
Part: collar_0
Overall 13.2 x 13.2 x 4.0
Volume 0.15 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



DETAIL C (section A-A, x 2 to 6, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

1. Ø5.0 shank x 6, Ø7.0 x 0.6 flange, Ø3.0 pin x 4.0 offset 0.9; 2 mm hex socket

the 60 — v12

Eccentric bush (x3)

Cadrane · sheet 10 of 13 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / brass (production)

Process: print stem down / pin up, 0.12 mm; the 1 mm flange ledge is the overall flange 7.0 x 10.6

mm. Tolerance ±0.1 unless stated; clearances: running 0.30,

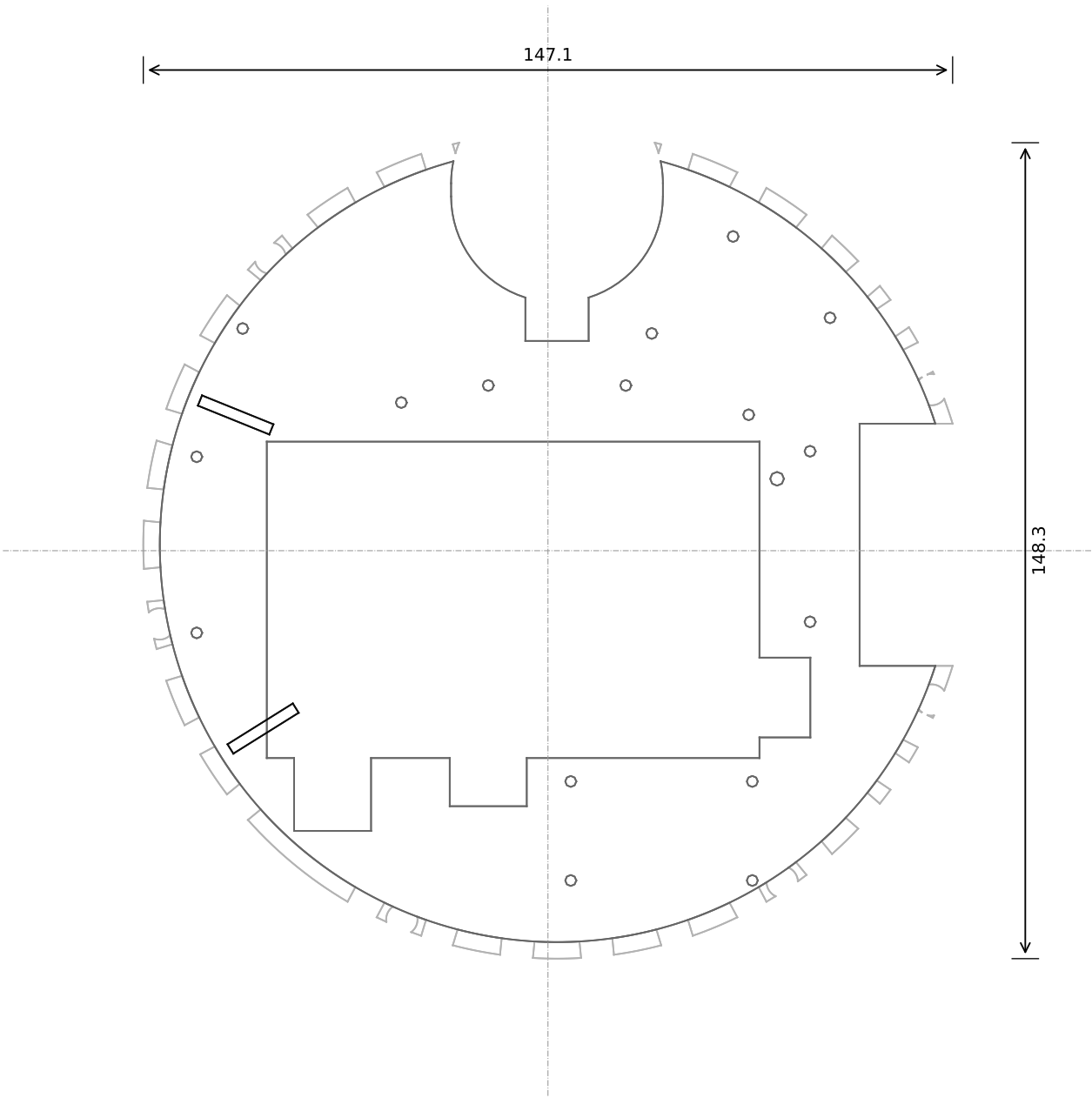
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: bush_0

Volume 0.16 cm³

Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 44 to 76, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

1. $r < 75.2 \times 8.0$: 4 web (z 4-8), 3 duct (z 1-4) closed by a 1 mm plate in a recess; rebate $r\ 72.2\text{-}75.2$, z 4-8 for the ring's flange, 7 x M3 csk from below at $r\ 73.7$
2. duct: collector $r\ 66\text{-}70.2$ (front and sides), straight channels 3 wide on a 4 pitch beside the Pi window, rear plenum $r\ 55+$, blower envelope $28 \times 18 \times 5$ at az 328 (not fitted); 2.5 walls round every through-cut
3. Pi window (Pi at (-52, -38), with the USB-C plug head, HDMI plug and USB-A plug notches), carriage hole, port slot; 16 x M2.5 tapped blind 3.0 deep from the top (converter audio, motion, servo frame, connect bracket, speaker cradle) and an M3 ground bond at (40.0, 12.0); 6 x M2.5 tapped from below for the closing plate (6082 aluminium, CNC two-sided, black hard anodised on every face) at the ground bond and under the converter
4. 2 ribs 2.0×6.0 tall on the top face at $r\ 56\text{-}70$, az [158, 212]

the 60 — v12

Base plate - aluminium core (the heatsink)

Cadrane · sheet 11 of 13 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: 6082 aluminium, CNC two-sided, black hard anodised on every face

Process: machined, not printed (print a PETG stand-in without the duct for to fit the 147.1 x 148.3 x 14.0

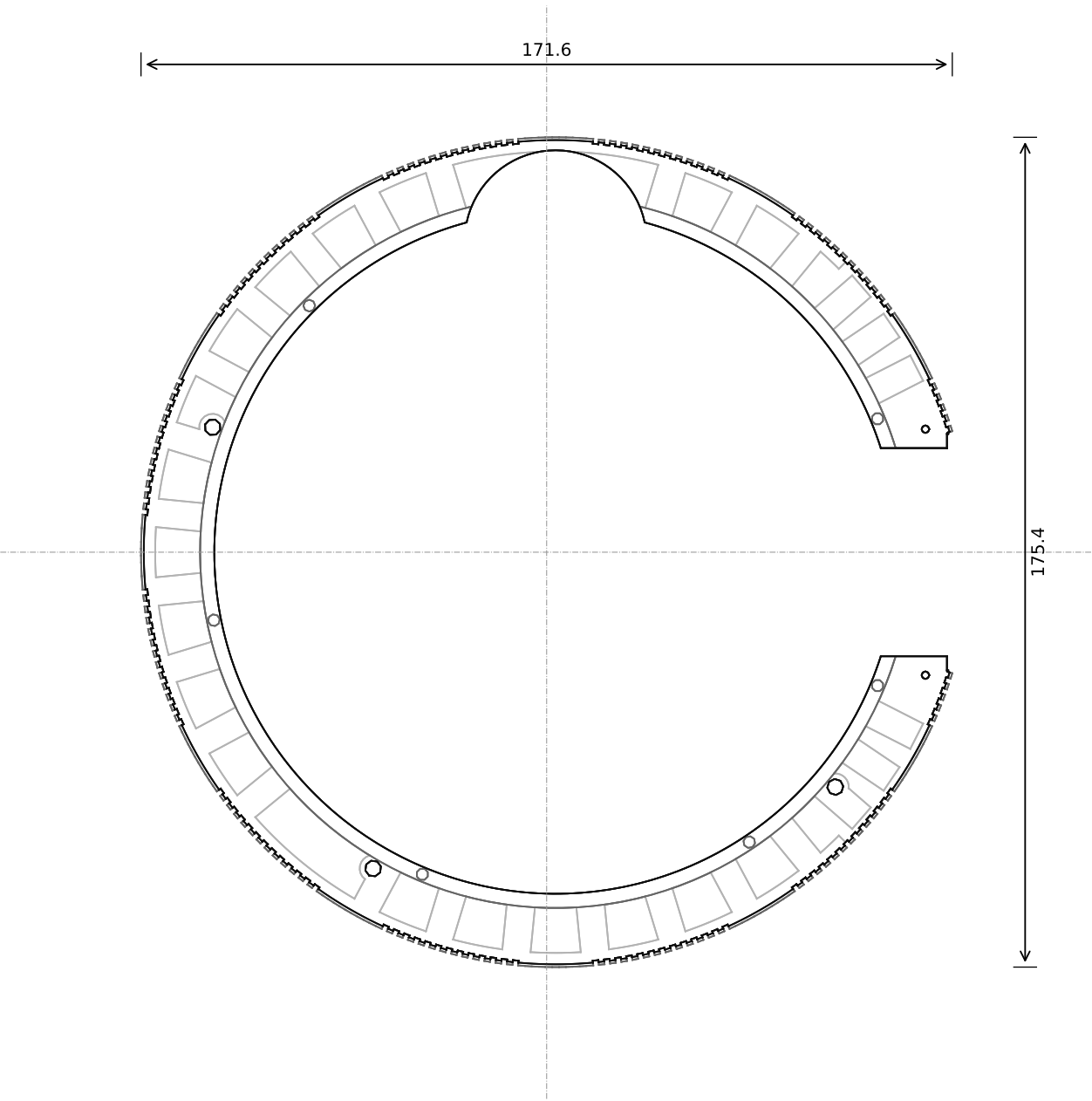
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,

static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Volume 51.35 cm³

Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



8.0

DETAIL C (section A-A, x 56 to 88, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- r 75.2-87.7 x 8.0; engine-turned edge: 12 plain blocks 8° wide at 0/30/60..., 156 square grooves 1.2 x 1.0 at 2.4 pitch between them; 1.0 chamfer top and bottom outer edges
- inward flange r 72.2-75.2, z 4-8, 7 x M3 tapped blind from below at r 73.7, az [22.5, 78.75, 135, 191.25, 247.5, 303.75, 337.5]
- intake: 1.2 undercut on the bottom outer edge outside r 84.7 over az 45-315, feeding 21 radial grooves 6.0 x 2.0 in the underside (the pad closes them); exhaust: 6 grooves at az [27.0, 34.0, 41.0, 319.0, 326.0, 333.0], opening downward where the pad is cut away
- 3 x M3 csk from below at r 77.2, az [160, 240, 320] (the structure's pillars); 2 x Ø1.6 tapped M2 at r 78.2, t ±26 (the port face's rail); port slot 44.0 wide

the 60 — v12

Rim ring (steel)

Cadrane · sheet 12 of 13 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: mild steel, machined, black (RULING 6 Sep: steel for mass)

Process: machined, not printed

mm. Tolerance ±0.1 unless stated; clearances: running 0.30,

static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

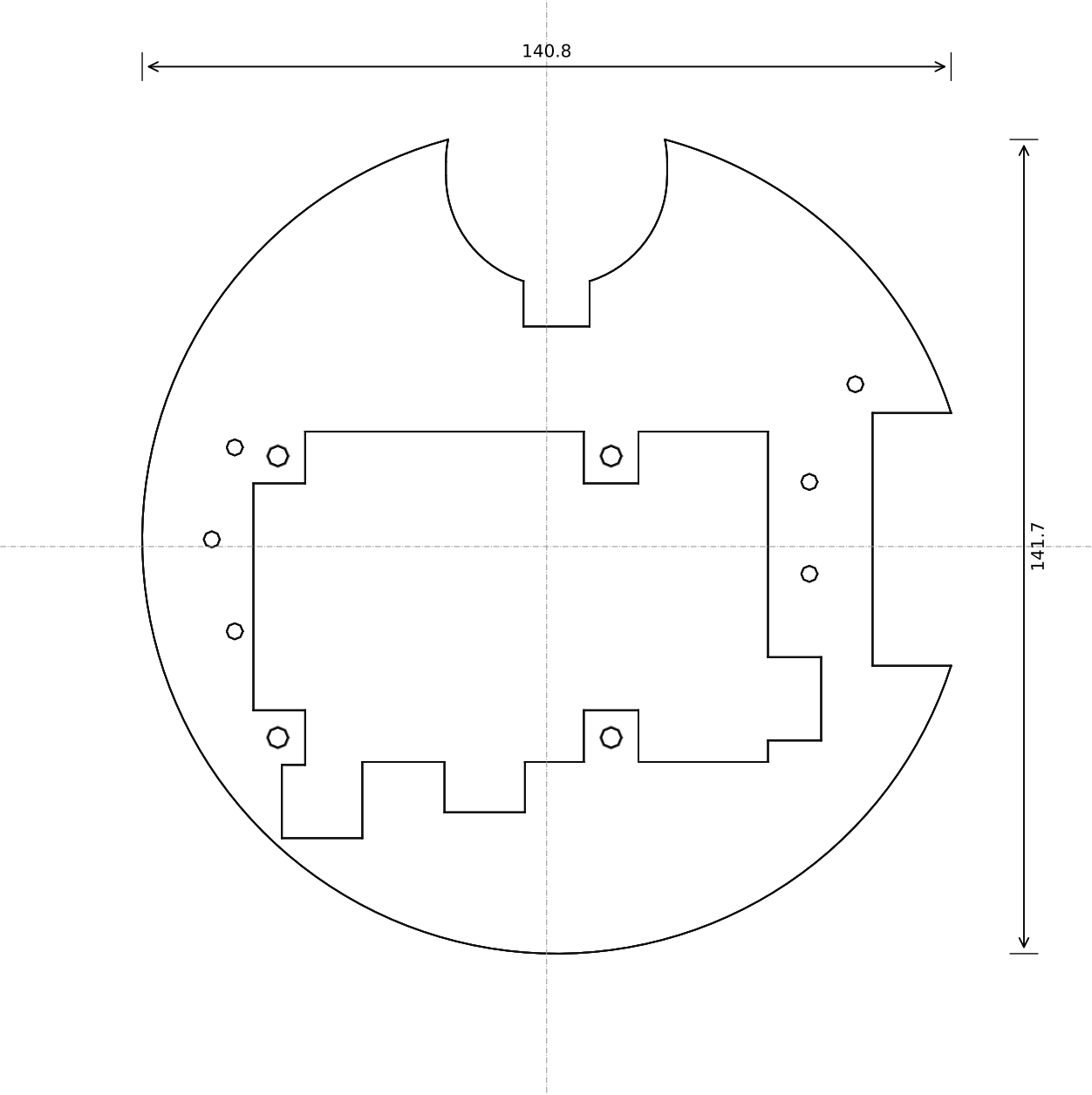
Part: rim_ring_STEEL

Overall 171.6 x 175.4 x 8.0

Volume 42.58 cm³

Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 40 to 72, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. disc r 72.1 x 1.0, flush in the core's recess; same Pi window, carriage hole and port slot as the core
- 2. four 9.5 square lands reaching into the Pi window at the Pi's holes [(-48.5, -34.5), (9.5, -34.5), (-48.5, 14.5), (9.5, 14.5)], Ø2.7 csk from below: the Pi and adapter stack screws to them (M2.5 x 6 into the standoffs)
- 3. 6 x M2.5 countersunk; unscrew to clean the fin channels

the 60 — v12

Closing plate (aluminium)

Cadrane · sheet 13 of 13 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: 1 mm aluminium, laser-cut, black anodised
Process: laser-cut, not printed
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: closing_plate_AL
Overall 140.8 x 141.7 x 1.0
Volume 9.32 cm³
Scale: see each view (fit)